Notes: Mian, Rao, and Sufi (QJE, 2013)

Household Balance Sheets, Consumption, and the Economic Slump

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1 Big picture

- **Question.** How much did the 2006–2009 collapse in household housing wealth reduce consumption, and does the marginal propensity to consume (MPC) out of housing wealth differ across households by wealth and leverage?
- **Setting.** The paper exploits the enormous geographic dispersion in housing wealth losses across U.S. counties and ZIP codes during the Great Recession.
- **Core empirical object.** The key regressor is the *housing net worth shock*: the part of the change in household net worth from 2006 to 2009 that comes from the drop in home values.
- **Main risk-sharing result.** Consumption is *not* insulated from local wealth shocks. The elasticity of spending with respect to the housing net worth shock is about **0.6 to 0.8**, strongly rejecting full consumption risk-sharing.
- **Main level result.** The average MPC out of housing wealth is about **5 to 7 cents per dollar**. The response is much larger for automobiles and other durable goods than for groceries.
- **Main heterogeneity result.** The MPC is much higher in poorer and more levered places. ZIP codes with low income or high housing loan-to-value ratios cut spending much more for the same dollar decline in home values.
- **Credit-constraint result.** ZIP codes suffering larger housing wealth losses see sharper declines in home-equity limits, refinancing, and credit quality; these effects are especially strong in low-income and high-leverage areas.
- **Quantitative implication.** Using an MPC around 0.06, the aggregate drop in housing wealth implies a spending decline of roughly $336 billion, about **40%** of the observed shortfall in spending relative to pre-2006 trend.
- **Why the paper matters.** The paper links household balance sheets, debt, and local demand collapse in a direct way. Its central message is that the *distribution* of wealth losses matters, not just the aggregate loss itself.

2 Data and measurement (key objects)

2.1 Observed consumption panel

The paper builds new measures of spending that do *not* rely on self-reported household surveys.

- **ZIP-code automobile spending.** New auto purchases come from **R.L. Polk** registration data from 1998 to 2012. The data identify the purchaser's ZIP code and measure the quantity of new autos bought.
- **County spending.** County-level spending from 2005 to 2009 comes from **MasterCard Advisors**. The sample is based on a 5% random sample of all transactions processed by MasterCard and includes both credit and debit card purchases.
- **Spending categories.** The paper aggregates merchant-level transactions into three broad groups:

  - durable goods (furniture, appliances, home improvement),
  - groceries,
  - other nondurable goods (health-related purchases, gasoline, clothing, sporting/hobby, department stores, restaurants).

- **Scaling to total spending.** The MasterCard shares of spending across counties are used to allocate aggregate retail sales from the Census, so the dollar changes in spending can be compared directly to dollar changes in wealth.

2.2 Observed wealth panel

Household net worth in ZIP code i at date t is defined as

$$NW_i^t = S_i^t + B_i^t + H_i^t - D_i^t,$$

where S is stocks, B is bonds and deposits, H is housing wealth, and D is debt.

- **Financial wealth.** The paper uses IRS Statistics of Income (SOI) data on dividend and interest income to allocate aggregate household financial assets from the Federal Reserve Flow of Funds across ZIP codes.
- **Housing wealth.** Housing wealth is constructed from the 2000 Census as the number of homeowners times median home value, projected forward using the CoreLogic ZIP-code house price index and estimates of changes in homeownership and population.
- **Debt.** Household debt comes from Equifax Predictive Services at the ZIP-code level.

The change in net worth from 2006 to 2009 is decomposed into financial and housing components:

$$\Delta NW_i^{06-09} = \Delta \log p_{06-09}^S \cdot S_i^{2006} + \Delta \log p_{06-09}^B \cdot B_i^{2006} + \Delta \log p_{06-09}^{H,i} \cdot H_i^{2006}.$$

The paper defines the **financial net worth shock** as

$$FNS_i = \frac{\Delta \log p_{06-09}^S \cdot S_i^{2006} + \Delta \log p_{06-09}^B \cdot B_i^{2006}}{NW_i^{2006}},$$

and the **housing net worth shock** as

$$HNS_i = \frac{\Delta \log p_{06-09}^{H,i} \cdot H_i^{2006}}{NW_i^{2006}}.$$

- **Interpretation.** The main source of cross-sectional variation is the housing component, not the financial component.
- **Important measurement assumption.** Financial wealth changes are measured using aggregate stock and bond indices, so the paper likely understates the true cross-sectional heterogeneity in financial-wealth shocks.

2.3 Other observed variables

- **House prices.** CoreLogic ZIP-code house price index.
- **Employment shares.** County Business Patterns data on construction, tradable, nontradable, and other sector employment shares.
- **Income.** IRS SOI income per household and ZIP-code adjusted gross income.
- **Credit conditions.** Equifax measures of home-equity limits, credit card limits, refinancing activity, and the fraction of borrowers with credit score below 660.
- **Leverage.** The housing leverage ratio, or housing loan-to-value ratio, is total mortgage plus home-equity debt divided by the value of owner-occupied housing in the ZIP code.
- **Instrument.** Housing supply elasticity from Saiz (2010).

2.4 Summary statistics, sample, and timing

- The county sample contains **944 counties**, covering about **82.1%** of the U.S. population in 2006.
- The population-weighted average housing net worth shock is about -9.2%; the average financial net worth shock is about -9.4%. So total household net worth falls by about **18.6%** on average.
- The population-weighted average drop in home value per household is about **$47,500**.
- The population-weighted average decline in total spending is about **9.2%**, or about **$3,400** per household.
- The spending collapse is concentrated in autos and other durables.

3 Models and empirical specifications (all equations + interpretation)

3.1 Benchmark consumption insurance equation

The representative-agent or full risk-sharing benchmark is

$$\Delta \log C_{it} = \alpha_t + \eta \, \Delta \log X_{it} + \varepsilon_{it}, \tag{1}$$

where C_{it} is household consumption and X_{it} is wealth.

- **Testable prediction.** Under full insurance, $\eta = 0$: idiosyncratic changes in wealth should not move idiosyncratic consumption growth.
- **Interpretation.** This is the paper's simplest test against representative-agent risk-sharing logic.

3.2 Heterogeneous MPC under limited risk-sharing

To study whether the spending response depends on initial net worth, the paper considers

$$\Delta C_{it} = \alpha_t + \beta_1 \, \Delta NW_{it} + \beta_2 \, NW_{i,t-1} + \beta_3 \, \Delta NW_{it} \times NW_{i,t-1} + \varepsilon_{it}. \tag{2}$$

- **Precautionary-savings interpretation.** In Carroll–Kimball type models, the consumption function is concave in wealth, implying $\beta_3 < 0$: low-wealth households have a higher MPC.
- **Credit-constraint interpretation.** The same sign restriction emerges if lower net worth means tighter borrowing constraints.
- **Core idea.** The response of spending to a wealth shock should depend on who loses the wealth.

3.3 Housing leverage, debt, and macro amplification

The paper's conceptual point is that leverage both magnifies losses on the balance sheet of debtors and concentrates those losses on households with high MPCs.

- For a given aggregate decline in housing wealth, more leverage implies a bigger contraction in spending if debtors have higher MPCs than creditors.
- This logic connects household debt to demand-driven recessions and to the broader "debt deflation" narrative.

3.4 County-level elasticity regression

The main county-level reduced-form specification relates spending growth to the housing net worth shock:

$$\Delta \log \text{Spend}_c^{06-09} = \alpha + \gamma\, HNS_c + \lambda\, FNS_c + X_c'\delta + u_c. \tag{3}$$

- **Dependent variable.** Growth in county spending from 2006 to 2009.
- **Main regressor.** The housing net worth shock, HNS_c.
- **Controls.** Financial net worth shock, county employment shares by sector, income, and net worth.
- **Weighting and inference.** Regressions are weighted by the number of households and standard errors are clustered at the state level.
- **Interpretation.** The coefficient on HNS_c is an elasticity: the percent change in consumption associated with a one-unit change in the housing-wealth-driven component of net worth.

3.5 County-level MPC regression in dollars

The paper also estimates the spending response in levels:

$$\Delta\text{Spend}_c^{06-09} = \alpha + \theta_1\, \Delta\text{HomeValue}_c^{06-09} + X_c'\delta + e_c. \tag{4}$$

A nonlinear variant allows

$$\Delta\text{Spend}_c^{06-09} = \alpha + \theta_1\, \Delta\text{HomeValue}_c^{06-09} + \theta_2 \left(\Delta\text{HomeValue}_c^{06-09}\right)^2 + X_c'\delta + e_c. \tag{5}$$

- **Interpretation.** The coefficient on the change in home value is the average MPC out of housing wealth.

- **Nonlinearity.** The quadratic term tests whether the MPC varies with the size of the housing wealth loss.

3.6 ZIP-code heterogeneity by wealth and income

Because wealth dispersion is much larger within counties than across counties, the paper turns to ZIP-code data and estimates

$$\Delta\text{AutoSpend}_z^{06-09} = \alpha + \theta_1\,\Delta\text{HomeValue}_z^{06-09} + \theta_2\,W_z^{2006} + \theta_3\,\Delta\text{HomeValue}_z^{06-09} \times W_z^{2006} + e_z, \quad (6)$$

where W_z^{2006} is either net worth per household or income per household.

- **Interpretation.** A negative interaction term means richer places have a smaller MPC out of housing wealth.
- **Why ZIP codes?** Variation in wealth is much larger at the ZIP-code level because households sort across neighborhoods.

3.7 ZIP-code heterogeneity by leverage

The leverage specification is

$$\Delta\text{AutoSpend}_z^{06-09} = \alpha + \phi_1\,\Delta\text{HomeValue}_z^{06-09} + \phi_2\,LTV_z^{2006} + \phi_3\,\Delta\text{HomeValue}_z^{06-09} \times LTV_z^{2006} + e_z. \quad (7)$$

- **Interpretation.** A positive interaction term means more levered households cut spending more for the same dollar decline in home values.
- **Debt-deflation logic.** This is the empirical implementation of Fisher's idea that debt amplifies the real consequences of asset-price declines.

3.8 Credit-constraint regressions

To assess whether the spending response operates through collateral and credit conditions, the paper estimates

$$\Delta\text{CreditOutcome}_z = \alpha + \lambda_1\,\Delta\text{HomeValue}_z + \lambda_2\,LTV_z + \lambda_3\,\Delta\text{HomeValue}_z \times LTV_z + \lambda_4\,Inc_z + \lambda_5\,\Delta\text{HomeValue}_z \times Inc_z + \varepsilon \quad (8)$$

The outcomes are:

- change in home-equity limits,
- change in credit card limits,
- change in refinancing volume,
- change in the fraction of subprime borrowers.

- **Interpretation.** These regressions ask whether falling house prices tighten borrowing conditions, and whether that tightening is stronger in low-income or high-leverage areas.

4 Identification and instruments (what makes the estimates credible)

4.1 What fails in a naive interpretation

A raw negative correlation between local house prices and local spending could reflect many confounds:

- local labor-demand shocks,
- construction-specific downturns,
- cross-county differences in long-run income trends,
- broader local recession exposure unrelated to household balance sheets.

So the paper needs a source of variation that shifts local housing wealth independently of these confounds.

4.2 Why housing supply elasticity is the key instrument

The paper uses the Saiz (2010) housing supply elasticity measure as an instrument for exposure to the housing boom-bust cycle.

- More inelastic areas experienced larger price booms from 2002 to 2006 when mortgage credit expanded nationwide.
- Those same places also built up more leverage because rising home values supported more borrowing.
- When the boom reversed, these areas experienced larger declines in housing net worth from 2006 to 2009.

Formally, the first stage is of the type

$$HNS_c = \alpha + \pi \text{ HousingSupplyElasticity}_c + \nu_c. \tag{9}$$

Because higher elasticity means easier supply, the coefficient is positive when the dependent variable is the housing net worth shock measured as a percentage change in total net worth. Inelastic places (low elasticity) therefore have *more negative* shocks.

4.3 Why the instrument is credible

The paper shows that housing supply elasticity is not systematically related to several alternative channels:

- no differential permanent-income shock, proxied by wage growth,
- no differential exposure to the construction sector in 2006,
- no differential construction employment growth from 2002 to 2006,
- only slightly slower population growth in inelastic places.

This matters because it weakens the idea that the spending collapse is just picking up an omitted construction shock or other industry-specific downturn.

4.4 What the IV estimate captures

The IV estimate is *inclusive* of several channels:

- the direct wealth effect of lower home values,
- the collateral and credit-constraint channel,
- local general-equilibrium feedback through reduced nontradable employment and income.

So the paper is explicit that its estimate is not a narrow partial-equilibrium wealth effect. It is the overall causal effect of the local housing boom-bust cycle on local consumption.

4.5 Why ZIP-code analysis strengthens the heterogeneity results

County-level cross-sections have limited variation in wealth across places. ZIP codes provide much richer within-county variation in wealth and leverage, which greatly improves statistical power for estimating heterogeneous MPCs.

4.6 Relation to the literature

- Against representative-agent complete-market models, the paper shows strong failure of local consumption insurance.
- Relative to the housing-wealth-effect literature, the paper brings unusually direct micro-level consumption and balance-sheet data to the Great Recession.
- Relative to debt-deflation and household-finance macro models, it provides direct evidence that leverage changes the size of the spending response to asset-price shocks.

5 Tables and figures

5.1 Figure I and Table I: national wealth shocks and summary statistics

Read:

- Figure I shows the national paths of asset prices after 2006: house prices fall about 30%, stock prices crash and partially recover, and bond prices rise sharply.
- Table I shows that the average county experiences a large housing wealth shock and a similarly sized financial wealth shock, but cross-sectional variation is overwhelmingly driven by housing.
- The population-weighted average decline in home value is about $47,500 per household, and the spending drop is concentrated in autos and durables.
- This is why the paper centers the analysis on local housing net worth shocks rather than financial wealth shocks.

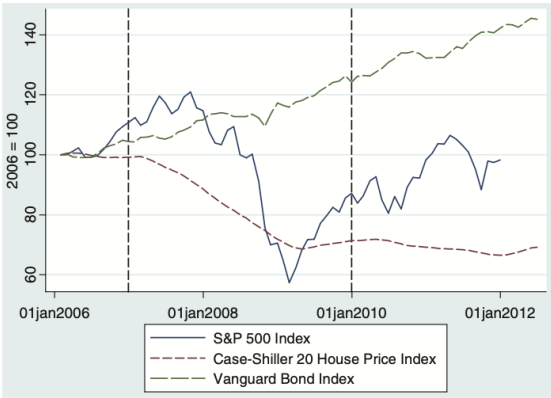

FIGURE I

Wealth Shocks during the Great Recession

This figure plots returns on the S&P 500, the Case-Shiller 20 MSA house price index, and the Vanguard Bond Index. All three indices are scaled to be 100 at the beginning of 2006. The dotted lines represent the end of years 2006 and 2009.

TABLE I

Summary Statistics

	N	Mean	Std. dev.	10th	90th	Weighted mean	Weighted std. dev.
Housing net worth shock, 2006–9	944	−0.063	0.083	−0.169	0.003	−0.092	0.097
Financial net worth shock, 2006–9	944	−0.096	0.011	−0.108	−0.084	−0.094	0.010
Change in home value, $000, 2006–9	944	−28.4	38.4	−79.1	1.2	−47.5	49.1
Spending growth, 2006–9	944	−0.059	0.135	−0.229	0.110	−0.092	0.113
Change in spending, $000, 2006–9	944	−1.7	4.6	−6.7	3.3	−3.4	4.4
Change in auto spending, $000, 2006–9	944	−2.6	1.6	−4.5	−1.0	−3.3	2.0
Change in other durables spending, $000, 2006–9	944	−0.6	1.3	−2.0	0.5	−1.1	1.1
Change in grocery spending, $000, 2006–9	944	0.5	0.9	−0.2	1.5	0.5	0.7
Change in other non-durable spending, $000, 2006–9	944	1.0	2.8	−1.6	4.0	0.5	2.4
Employment share in construction, 2006	944	0.119	0.054	0.065	0.182	0.125	0.048
Employment share in tradables, 2006	944	0.130	0.102	0.032	0.247	0.110	0.071
Employment share in other, 2006	944	0.522	0.232	0.274	0.830	0.667	0.268
Employment share in nontradables, 2006	944	0.210	0.067	0.137	0.283	0.216	0.051
Income per household, $000, 2006	944	52.2	15.9	38.2	70.2	59.9	18.9
Net worth per household, $000, 2006	944	429.9	246.7	230.5	684.5	520.8	288.8
Housing leverage ratio, 2006	944	0.616	0.229	0.360	0.902	0.608	0.179
Housing supply elasticity, Saiz	540	2.192	1.044	0.943	3.589	1.715	0.968
Number of households, thousands	944	98.2	187.5	12.8	237.8	455.9	666.2
Change in home equity limit, $000, 200–9	944	−0.473	4.786	−3.323	2.857	−0.725	3.637
Change in credit card limit, $000, 2006–9	944	−1.043	2.419	−3.567	1.778	−1.574	1.781
Change in fraction of subprime borrowers, 2006–9	944	−0.010	0.024	−0.038	0.019	−0.004	0.024
Change in refinancings, $000, 2006–9	944	1.236	6.646	−5.247	7.263	−1.268	8.518

Notes. This table presents summary statistics for the counties in our sample. The sample is restricted to 944 counties for which we have data on the value of housing stock. These counties represent 82.1% of total U.S. population in 2006. The housing net worth shock reflects the growth in total net worth due to the growth in housing net worth. The financial net worth shock reflects growth in total net worth due to growth in financial net worth. The housing net worth shock and the financial net worth shock sum up to the growth in total net worth. Other durables include purchases at furniture, home appliance, and home center stores. Other nondurables include purchases at health, gasoline, clothing, hobby and sporting, and department stores. See the text for the corresponding NAICS codes.

5.2 Figure II and Table II: the source of identifying variation

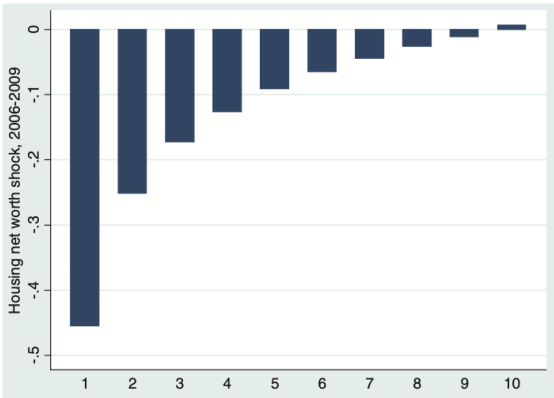

FIGURE II

Housing Net Worth Shock by Decile

This figure sorts ZIP codes into deciles (weighted by population) based on the housing net worth shock and shows each decile's housing net worth shock. The housing net worth shock reflects the growth in total net worth per household due to the growth in housing net worth.

TABLE II

Housing Supply Elasticity as a Source of Variation

		Housing supply elasticity	Constant	N	R^2
(1)	Housing net worth shock, 2006–9	0.046**	−0.174**	540	0.190
		(0.011)	(0.037)		
(2)	Change in home value, $000, 2006–9	27.795**	−95.740**	540	0.284
		(7.874)	(23.210)		
(3)	Change in wage growth (2002–6) – (1998–2002)	−0.002	−0.010	540	0.002
		(0.004)	(0.008)		
(4)	Employment share in construction, 2006	0.002	0.122**	540	0.003
		(0.003)	(0.008)		
(5)	Construction employment growth, 2002–6	0.005	0.940**	540	0.000
		(0.015)	(0.042)		
(6)	Population growth, 2002–6	0.012*	0.018	538	0.026
		(0.005)	(0.012)		
(7)	Income per household, 2006	−5.378**	69.392**	540	0.080
		(0.985)	(2.191)		
(8)	Net worth per household, 2006	−88.389**	674.620**	540	0.083
		(20.689)	(47.965)		

Notes. This table presents coefficients from county-level univariate regressions regressing variables on the housing supply elasticity instrument. Each row is a separate regression. The first two rows represent the first stage estimates of the housing net worth shock and the change in home value, respectively, on housing supply elasticity. Standard errors are heteroskedasticity robust, clustered at the state level. All regressions are weighted by the number of households in the county. **,* Coefficient statistically different than 0 at the 1% and 5% confidence levels, respectively.

Read:

- Figure II sorts ZIP codes by the housing net worth shock: the bottom decile loses almost **45%** of total net worth from housing, whereas the top decile sees almost no loss.
- Table II shows that housing supply elasticity strongly predicts both the housing net worth shock and the dollar decline in home values.
- The same table also shows little relationship between housing supply elasticity and wage growth or construction exposure, supporting the IV strategy.
- Inelastic places are richer and wealthier before the crash, but that level difference is differenced out in the main spending specifications.

5.3 Figure III and Table III: elasticity of spending with respect to the housing net worth shock

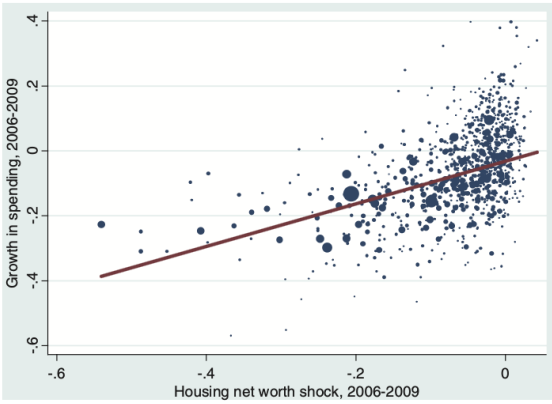

FIGURE III

Elasticity of Spending with Respect to Housing Net Worth Shock

The scatterplot relates total spending growth in a county from 2006 to 2009 to the housing net worth shock over the same time period. The housing net worth shock reflects the growth in total net worth per household due to the growth in housing net worth. The scatterplot and regression line are weighted by the number of households in the county.

TABLE III

Net Worth Shock and Consumption Growth, 2006–9

	(1)	(2)	(3)	(4) IV	(5) State FE	(6) Excluding AZ, CA, FL, NV
Housing net worth shock, 2006–9	0.634**	0.613**	0.590**	0.774**	0.457**	0.869**
	(0.125)	(0.122)	(0.130)	(0.239)	(0.101)	(0.148)
Financial net worth shock, 2006–9		−0.595				
		(1.032)				
Construction employment share, 2006			−0.448**	−0.287	−0.171	−0.288
			(0.150)	(0.216)	(0.127)	(0.160)
Tradable employment share, 2006			0.051	0.011	0.042	−0.027
			(0.067)	(0.092)	(0.066)	(0.065)
Other employment share, 2006			−0.025	−0.045	−0.057	−0.058
			(0.038)	(0.050)	(0.037)	(0.039)
Nontradable employment share, 2006			0.193	0.095	0.228	0.106
			(0.157)	(0.167)	(0.137)	(0.158)
Ln(income per household, 2006)			−0.002	0.024	−0.006	0.028
			(0.033)	(0.047)	(0.046)	(0.045)
Ln(net worth per household, 2006)			−0.028	−0.035	−0.023	−0.034
			(0.018)	(0.023)	(0.020)	(0.025)
Constant	−0.034*	−0.092	0.167*	0.147	0.120	0.132
	(0.015)	(0.099)	(0.077)	(0.092)	(0.090)	(0.087)
N	944	944	944	540	944	833
R^2	0.298	0.301	0.355	0.319	0.547	0.230

Notes. Dependent variable: total spending growth, 2006–9 (%). This table presents coefficients from regressions relating spending growth to the housing net worth shock. The unit of observation is a county. The housing net worth shock reflects the growth in total net worth due to the growth in housing net worth. The financial net worth shock reflects growth in total net worth due to growth in financial net worth. The housing net worth shock and the financial net worth shock sum up to the growth in total net worth. Standard errors are heteroskedasticity robust, clustered at the state level. All regressions are weighted by the number of households in the county. **,* Coefficient statistically different than 0 at the 1% and 5% confidence levels, respectively.

Read:

- The county-level spending elasticity with respect to the housing net worth shock is about **0.63** in the baseline OLS regression.
- The coefficient remains around **0.59** after adding financial net worth shocks and sectoral controls.
- The IV estimate using housing supply elasticity is about **0.77**.
- This is the paper's clean rejection of the full risk-sharing benchmark.

5.4 Figure IV and Table IV: the average MPC

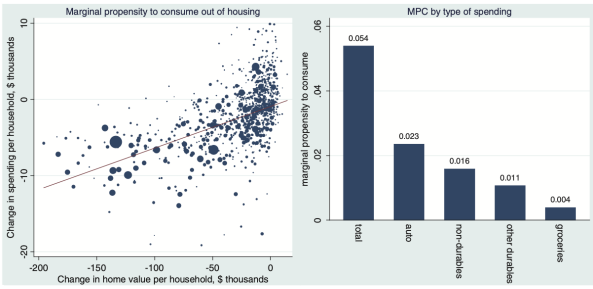

FIGURE IV

The Average Marginal Propensity to Consume

The left panel scatterplot relates the change in total spending per household in a county from 2006 to 2009 to the change in home values over the same time period. The scatterplot and regression line are weighted by the number of households in the county. The gradient of the line represents the average marginal propensity to consume. The right panel plots the marginal propensity to consume for various spending categories.

TABLE IV

Average Marginal Propensity to Consume out of Housing Wealth

	(1)	(2)	(3)	(4) IV	(5) State FE	(6) Excluding AZ, CA, FL, NV
Change in home value, $000, 2006–9	0.054**	0.119**	0.051**	0.072**	0.051**	0.094**
	(0.009)	(0.015)	(0.011)	(0.021)	(0.013)	(0.017)
(Change in home value, $, 2006–9)^2		0.432**				
		(0.076)				
Construction employment share, 2006			−9.748	−2.915	−7.449	−2.305
			(5.479)	(7.800)	(5.379)	(5.818)
Tradable employment share, 2006			2.034	0.438	1.516	−0.795
			(2.235)	(3.783)	(2.190)	(2.496)
Other employment share, 2006			−1.568	−3.037	−2.186	−2.629
			(1.459)	(1.850)	(1.418)	(1.466)
Nontradable employment share, 2006			−1.797	−3.256	−3.341	−4.106
			(5.438)	(5.983)	(5.048)	(5.349)
Income per household, $000, 2006			−0.056*	−0.019	−0.043	−0.022
			(0.023)	(0.032)	(0.030)	(0.029)
Net worth per household, $000, 2006			0.003*	0.002	0.002	0.002
			(0.001)	(0.001)	(0.002)	(0.001)
Constant	−0.830	0.263	3.311**	3.211**	3.396**	3.415**
	(0.536)	(0.554)	(0.678)	(0.928)	(0.861)	(0.837)
N	944	944	944	540	944	833
R^2	0.362	0.423	0.421	0.347	0.573	0.336

Notes. Dependent variable: change in spending 2006–9 ($000). This table presents coefficients from regressions relating the change in household spending to the change in home value between 2006 and 2009. Both the change variables are in thousands of dollars. All regressions are at the county level. Standard errors are heteroskedasticity robust, clustered at the state level. All regressions are weighted by the number of households in the county. **,* Coefficient statistically different than 0 at the 1% and 5% confidence levels, respectively.

Read:

- In dollar levels, the average MPC out of housing wealth is about **5.4 cents** in OLS and about **7.2 cents** in IV.

- Excluding the four biggest boom-bust states (Arizona, California, Florida, Nevada) raises the estimate to about **9.4 cents**, reflecting nonlinearities.
- The spending response is strongest for autos and other durables and weakest for groceries.
- These estimates imply that housing wealth losses explain a large share of the aggregate spending collapse.

5.5 Table V and Figure V: poorer households have higher MPCs

TABLE V

HETEROGENEITY IN THE MPC BY WEALTH AND INCOME

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Dependent variable: ΔTotal Spending ($000), 2006–9		*Dependent variable: ΔAuto Spending ($000) 2006–9*		*Dependent variable: ΔAuto Spending ($000) 2006–9*		
	County-level analysis		County-level analysis		ZIP code–level analysis		
Δ Home value, $000, 2006–9	0.076**	0.065**	0.034**	0.047**	0.018**	0.023**	0.025**
	(0.012)	(0.015)	(0.005)	(0.005)	(0.001)	(0.002)	(0.002)
Net worth, $millions, 2006	−4.289*		−1.81**			−0.354	
	(2.132)		(0.665)			(0.243)	
(Δ Home value)*(Net worth, 2006)	−0.038		−0.024*			−0.007**	
	(0.024)		(0.009)			(0.001)	
Income per household, $ millions, 2006		−64.042*		−31.814**			−4.020
		(28.158)		(7.819)			(3.136)
(Δ Home value)*(Income per household, 2006)		−0.180		−0.432**			−0.095**
		(0.332)		(0.100)			(0.022)
Constant	1.247	2.829*	−1.30**	−0.361	−2.075**	−1.883**	−1.809**
	(0.679)	(1.212)	(0.20)	(0.332)	(0.170)	(0.121)	(0.117)
N	944	944	944	944	6,263	6,220	6,263
R^2	0.462	0.478	0.427	0.440	0.153	0.161	0.163

Notes. This table presents coefficients from regressions relating the change in household spending to the change in home value between 2006 and 2009. Regressions in columns (1) through (4) are at the county level, and regressions in columns (5) through (7) are at the ZIP code level. The dependent variables is the change in total spending in columns (1) and (2), and the change in spending on autos in columns (3) through (7). Throughout, Δ signifies change in thousands of dollars. Standard errors are heteroskedasticity robust, clustered at the state level. All regressions are weighted by the number of households. **,* Coefficient statistically different than 0 at the 1% and 5% confidence levels, respectively.

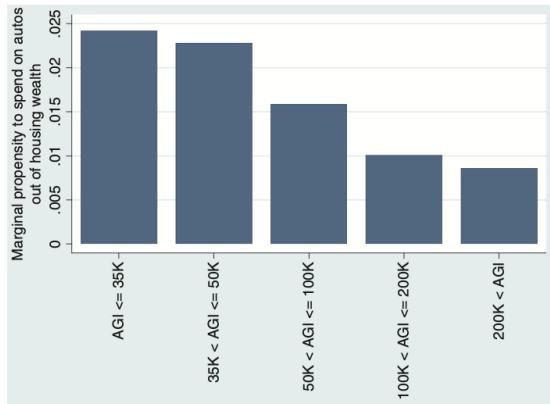

FIGURE V

Marginal Propensity to Consume out of Housing across 2006 Income Distribution

Read:

- County-level interaction estimates are noisy because wealth variation across counties is limited.
- ZIP-code regressions are much more informative: the average ZIP-level auto MPC is about **1.8 cents**, and the interaction with income or net worth is significantly negative.
- Figure V shows the magnitude: ZIP codes with average income below $35,000 have an MPC that is about **three times** as large as ZIP codes with income above $200,000.
- The distribution of losses therefore matters as much as the aggregate loss itself.

5.6 Table VI and Figure VI: levered households have higher MPCs

Read:

- Housing leverage is essentially orthogonal to ZIP-code income and net worth, so leverage is a separate source of heterogeneity.
- The interaction between home-value changes and leverage is strongly positive: more levered ZIP codes cut auto spending more for the same dollar decline in home values.
- Figure VI shows the magnitude clearly: ZIP codes with housing leverage below 30% have an MPC around **1 cent**, while ZIP codes above 90% have an MPC around **3 cents** for autos.
- This is the empirical content of the debt-deflation channel in the paper.

TABLE VI

Heterogeneity in the MPC: The Role of Housing Debt

	(1)	(2)	(3)	(4)	(5)
	Dependent variable: Housing leverage ratio, 2006		*Dependent variable: ΔAuto spending ($000), 2006–9*		
Δ Home value, $000, 2006–9			0.006**	0.010**	0.011**
			(0.002)	(0.002)	(0.002)
Housing leverage ratio, 2006			−2.112**	−2.146**	−2.191**
			(0.228)	(0.232)	(0.230)
(Δ Home value)*(Housing leverage ratio, 2006)			0.021**	0.020**	0.020**
			(0.003)	(0.004)	(0.003)
Net worth, $millions, 2006	0.004			−0.153	
	(0.013)			(0.158)	
(Δ Home value)*(Net worth, 2006)				−0.005**	
				(0.001)	
Income per household, $ millions, 2006		0.327			0.022
		(0.233)			(1.627)
(Δ Home value)*(Income per household, 2006)					−0.059**
					(0.015)
Constant	0.595**	0.576**	−0.786**	−0.667**	−0.705**
	(0.011)	(0.016)	(0.150)	(0.150)	(0.157)
N	6,385	6,448	6,222	6,182	6,222
R^2	0.000	0.003	0.272	0.272	0.279

Notes. This table presents coefficients from regressions relating the change in household spending to the change in home value between 2006 and 2009. All regressions are at the ZIP code level. The housing leverage ratio is defined to be the ratio of mortgage and home equity debt to home value in a ZIP code as of 2006. Throughout, Δ signifies change in thousands of dollars. Standard errors are heteroskedasticity robust, clustered at the state level. All regressions are weighted by the number of households in a ZIP code. **,* Coefficient statistically different than 0 at the 1% and 5% confidence levels, respectively.

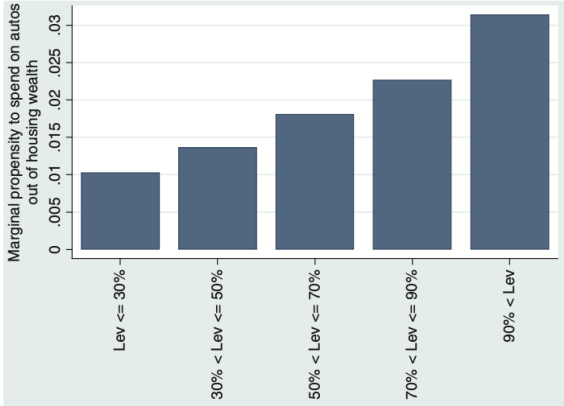

FIGURE VI

Marginal Propensity to Consume out of Housing across Housing Leverage Ratio Distribution

The figure plots the estimated marginal propensity to spend on autos based on the 2006 housing leverage ratio of a ZIP code. The housing leverage ratio is defined to be total mortgage and home equity debt scaled by the total value of owner-occupied homes in a ZIP code. The MPC is estimated using ZIP code–level data and regressing the change in spending on automobile purchases between 2006 and 2009 on the change in home values over the same period. Each regression is run separately for ZIP codes in a given housing leverage ratio category and the resulting MPC coefficient is plotted.

TABLE VII

Why Do Levered ZIP Codes Have a Higher MPC Out of Housing Wealth?

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
	Dependent variable: Δ Home equity limit, $000, 2006–9			*Dependent variable: Δ Credit card limit, $000, 2006–9*			*Dependent Variable: Δ Refinancing volume, $000, 2006–9*			*Dependent variable: Δ Fraction subprime borrowers, 2006–9*		
Δ Home value, $000, 2006–9	0.023**	−0.010*	0.040**	0.010**	0.009*	0.015**	0.119**	0.012	0.179**	−0.033**	−0.019**	−0.046**
	(0.002)	(0.004)	(0.006)	(0.001)	(0.004)	(0.003)	(0.007)	(0.015)	(0.019)	(0.002)	(0.003)	(0.002)
Housing leverage ratio, 2006		−1.234**			0.850*			13.378**			0.172	
		(0.429)			(0.390)			(0.857)			(0.267)	
(Δ Home value)*(Housing leverage ratio, 2006)		0.058**			0.002			0.179**			−0.024**	
		(0.007)			(0.007)			(0.019)			(0.004)	
Income per household, $ millions, 2006			17.295			−3.516			109.261**			0.543
			(11.630)			(2.224)			(40.775)			(1.677)
(Δ Home value)*(Income per household, 2006)			−0.187*			−0.070*			−0.536			0.165**
			(0.093)			(0.030)			(0.307)			(0.018)
Constant	0.216	0.972**	−0.603	−1.080**	−1.606**	−0.856**	4.077**	−4.058**	−1.306	−1.623**	−1.733**	−1.751**
	(0.114)	(0.273)	(0.610)	(0.094)	(0.224)	(0.134)	(0.318)	(0.620)	(2.098)	(0.083)	(0.202)	(0.129)
N	6,273	6,236	6,273	6,262	6,228	6,262	6,212	6,191	6,212	6,262	6,215	6,262
R^2	0.051	0.098	0.090	0.021	0.023	0.023	0.311	0.361	0.532	0.293	0.311	0.322

Notes. This table presents coefficients from regressions relating borrowing constraints to the change in home value between 2006 and 2009. All regressions are at the ZIP code level. The housing leverage ratio is defined to be the ratio of mortgage and home equity debt to home value in a ZIP code as of 2006. Throughout, Δ signifies change in thousands of dollars. Standard errors are heteroskedasticity robust, clustered at the state level. All regressions are weighted by the number of households in the ZIP code. **,* Coefficient statistically different than 0 at the 1% and 5% confidence levels, respectively.

5.7 Table VII: why leverage matters — tighter credit constraints

Read:

- Falling home values tighten credit conditions: home-equity limits fall, credit card limits fall, refinancing collapses, and the share of subprime borrowers rises.
- These effects are stronger in high-leverage ZIP codes for home-equity borrowing, refinancing, and credit scores.
- The leverage interaction is not important for credit card limits, suggesting that the key channel is specifically housing-collateral-based credit.
- Poorer ZIP codes also face larger tightening for a given decline in home values, so income and leverage independently amplify the spending response.

6 Short summary

- **Conclusion.** The collapse in household housing wealth during the Great Recession caused a large drop in consumption.
- **Identification insight.** Cross-sectional exposure to the housing boom-bust cycle is instrumented by housing supply elasticity, which shifts how strongly local demand translated into earlier house-price appreciation and later collapse.
- **Main quantitative lesson.** The average MPC out of housing wealth is around 5–7 cents, but it is much higher for poorer and more levered households.
- **Economic message.** Debt changes the macro effect of asset-price declines because it concentrates losses on households with the highest MPCs.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The representative-agent risk-sharing benchmark performs poorly during the Great Recession.
- Local household spending responds strongly to local housing net worth losses.
- The spending response is highly heterogeneous across the wealth and leverage distributions.
- The housing collateral channel appears central: larger home-value declines tighten borrowing constraints the most in poorer and more levered areas.

7.2 Economic intuition

A fall in home values is not just a paper loss. It reduces household net worth, weakens collateral, tightens credit access, and lowers the ability to refinance. Because households with low wealth and high leverage have the highest MPCs, they cut spending the most when they are hit. That is why the same aggregate decline in housing wealth can have very different aggregate consequences depending on who bears the losses.

7.3 Takeaway

This paper's central contribution is to show that the Great Recession was not simply a story about a fall in aggregate wealth. It was a story about *where* the losses occurred on household balance sheets. Housing debt amplified the real effects of the crash by concentrating losses on households most likely to reduce spending. That insight is central for thinking about macro fluctuations, household finance, and the design of stabilization policy.